

Credit Risk & Loan Approval Project

Business Review: Project Overview, Predictive Model, Portfolio Shortcomings & Recommendations

1. What This Project Is

This project builds and evaluates a **loan default prediction system** for a lending institution that has processed **255,000+ loan applications** totaling a **■32.58 Billion** disbursed portfolio, with an overall approval rate of **88.39%** (226K approved / 30K rejected). The work spans the full journey from raw applicant data to a deployable risk-screening model, and sits alongside two executive dashboards that describe how the institution is currently lending.

The pipeline cleans applicant records, engineers a set of affordability and risk features (EMI burden, disposable income after EMI, loan burden, credit-score tiers, income tiers, employment stability), profiles those features against actual loan outcomes, and trains a battery of classification models to predict which applicants are likely to **default** on repayment. The end deliverable is a production-style scoring pipeline — a fitted preprocessing step plus a trained classifier, saved and reloadable — intended to sit inside the underwriting workflow as a **pre-approval risk screen**.

In parallel, two dashboard-level analyses examine the institution's current lending patterns: an **Executive Dashboard** covering portfolio size, approval behavior, credit-score mix, income mix, and applicant demographics, and a **Loan & Finance Dashboard** focused specifically on debt burden, EMI, loan-to-income ratio, and default patterns by credit score. Together, the modeling work and the dashboards tell the same story from two angles: the institution currently approves almost everyone, and this project is the mechanism for tightening that up with an evidence-based, recall-oriented default filter.

2. The Predictive Model: Built to Catch Defaulters, Not Just to Be Accurate

Ten classification algorithms were screened on the cleaned, feature-engineered dataset, then the four strongest candidates were hyperparameter-tuned. The deciding metric throughout was **ROC-AUC** together with **recall on the default class** — not raw accuracy — because in a lending context the cost of the two error types is deeply asymmetric: approving a borrower who defaults costs the institution the full outstanding loan amount, while declining a borrower who would have repaid costs only the foregone interest income. A model optimized for accuracy alone will always be tempted to just approve everyone, since only 11.6% of the historical portfolio actually defaulted.

Why accuracy was explicitly rejected as the selection metric

The unweighted baseline screen made this concrete. Every tree-based model (Gradient Boosting, CatBoost, AdaBoost, XGBoost, Random Forest, Extra Trees) posted **88%+ accuracy** while catching only **2.5% to 8.4%** of actual defaulters — they were effectively predicting “no default” for nearly everyone and being rewarded for it by the class imbalance. Only **Logistic Regression, run with class-weighting to correct for the imbalance**, produced a model that actually identifies defaulters at a usable rate.

Model (weighted/tuned)	Accuracy	Precision	Recall (catches defaulters)	ROC-AUC
Logistic Regression (final model)	69.1%	22.8%	69.6%	0.7615
XGBoost (tuned, weighted)	69.3%	22.8%	68.7%	0.7604
CatBoost (tuned, weighted)	69.4%	22.9%	69.2%	0.7603

Model (weighted/tuned)	Accuracy	Precision	Recall (catches defaulters)	ROC-AUC
Random Forest (tuned, weighted)	87.3%	41.1%	21.5%	0.7471

Logistic Regression was selected as the final production model specifically because it matches the tuned tree ensembles on ROC-AUC while delivering far higher recall. Its final held-out performance: it correctly flags **70% of applicants who go on to default**, at the cost of a lower precision of **23%** — meaning roughly 3 of every 4 applicants it flags as high-risk would, in this historical data, have actually repaid.

What this tradeoff means in plain business terms

The model is deliberately biased toward **catching bad loans, even at the cost of some false alarms on good ones**. This makes it well-suited as a **screening / triage layer** that flags applicants for closer underwriter review or additional conditions (e.g. a co-signer, a smaller loan amount) rather than an automatic accept/reject switch. Used this way, its low precision is not a defect — a human or a secondary rule still makes the final call on the ~77% of flagged applicants who are actually creditworthy. Used as a blunt auto-decline gate, the same model would reject a large number of good borrowers and should not be deployed that way.

3. Business Patterns Observed — Shortcomings in the Current Lending Approach

The dashboard data describes how the institution is lending **today**, independent of the new model. Read against the modeling findings, several patterns stand out as genuine business risk, not just data curiosities.

3.1 Approval policy is running far more aggressively than the risk profile of the applicant pool

- The institution approves **88.39%** of all applicants — well above the typical 40–85% range seen across traditional banks, retail banks, and digital lenders/NBFCs — while the applicant pool itself skews toward weaker credit: the **Poor** credit-score tier alone accounts for roughly **115K applicants (about 45% of the pool)**, more than the Fair, Good, Very Good, and Excellent tiers combined.
- Among low-credit applicants specifically, the approval rate is **87.54%** — almost identical to the overall approval rate. In other words, having weak credit barely changes an applicant's odds of approval in current practice.
- **Problem:** approval decisions are not differentiating enough between low-risk and high-risk applicants at the point of decision. The institution is absorbing the full risk of a Poor-credit-heavy book without a correspondingly tighter approval filter.

3.2 Debt burden is high across the book, and it is being approved through rather than screened out

- **61.87%** of all applicants (158K) already carry a **High DTI** classification before taking on this loan, yet the dashboard shows most of them are still approved.
- Average **Debt-to-Income ratio is 0.50** across the whole book — the typical approved borrower is already committing roughly half their income to debt service before this new loan is added.
- The modeling work found the same pattern from a different angle: **DTIRatio on its own barely separates defaulters from non-defaulters**, and combinations like Low-DTI + High-Loan-Burden actually produced the single highest default rate observed (17.26%) — higher than the supposedly worse High-DTI + High-Burden cell (13.92%). This means the institution's apparent reliance on DTI as a risk gate is being undermined by a feature (Loan_Burden) that is a stronger driver of actual default and isn't being used as a hard gate today.
- **Problem:** the affordability signal that most reliably predicts default — loan burden and post-EMI disposable income, not raw DTI — is present in the data but is not evidently driving current approval decisions.

3.3 A meaningful share of the approved book cannot cover its own EMI from income

- Average EMI across the book is a manageable **■5.65K**, which looks healthy in aggregate — but this average hides a heavy-tailed distribution: modeled **Income_After_EMI** is negative for **more than a quarter of the portfolio**, with the worst case at roughly **–■21,800** per month, and EMI-to-income ratios run as high as **18×** monthly income for some borrowers.
- **Problem:** aggregate KPIs (average EMI, average DTI) look reassuring precisely because they average away a sizeable, genuinely unaffordable tail of the book — the population most likely to default is invisible in the top-line dashboard numbers.

3.4 Underwriting appears to lean on income and relationship signals, but not consistently or as a hard rule

- Loan-to-Income ratio is clearly influencing rejection in practice — rejected applicants average **~3.2** vs. **~2.0** for approved applicants — and the modeling confirms this is a real signal: **No-cosigner + High-loan-amount**

applicants default at 15.42%, nearly double the safest combination (**Co-signer + Low-loan-amount at 8.20%**).

- Employment type shows a real, usable spread — **Unemployed applicants taking Business loans default at 14.37%**, versus **8.05% for Full-time applicants taking Home loans** — but the dashboard's own employment-stability view shows roughly flat approval counts across employment-stability groups, suggesting this specific high-risk combination is not being priced or gated distinctly today.
- **Problem:** the individual signals the institution needs (co-signer status, loan size, employment type, loan purpose) are already being captured, but they are not being combined into the specific high-risk combinations that the data shows matter most.

3.5 Portfolio risk cannot currently be measured, only inferred

- Both dashboards report volume and approval metrics (total disbursed, approval rate, average interest rate) but neither reports **default rate, delinquency buckets, non-performing assets, portfolio-at-risk, expected credit loss, or recovery rate**.
- **Problem:** without these outcome metrics sitting next to the approval metrics, management can see how much has been lent and how eagerly, but not how much of it is actually being repaid on schedule — the dashboards currently measure growth, not portfolio health.

3.6 An unexplained, sharp drop in application volume

- Monthly applications run a steady 29K–33K from January through May, then fall sharply to 16K–17K from June onward and stay there — roughly a 50% drop that persists rather than reverting, which rules out a one-off blip.
- **Problem:** this is a material, sustained change in loan demand with no attributed cause in the data reviewed — left uninvestigated, the institution cannot tell whether this reflects a policy change, a marketing/channel change, or a genuine shift in market demand, each of which calls for a different response.

4. Solutions, Mapped to Each Problem

The recommendations below are ordered to match Section 3 and are written to be actionable with what the institution already has — the engineered features and the trained model — rather than requiring new data collection.

4.1 Tighten approval policy using risk-adjusted rules, not credit score in isolation

- Deploy the recall-focused Logistic Regression model as a **mandatory pre-approval screen**: every application is scored, and applications flagged high-risk are routed to enhanced underwriter review rather than being auto-approved by default. Given the model's 70% recall, this alone would bring the majority of future defaulters into a human review step before disbursement.
- Because model precision is only 23%, pair the score with a human-reviewed decision, not an auto-decline — the goal is to **redirect** risky applications toward added conditions (smaller loan, co-signer, shorter term) rather than blanket rejection, preserving approval volume on the ~77% of flagged applicants who are actually good credit risks.

4.2 Replace DTI as the primary debt-burden gate with Loan_Burden and Income_After_EMI

- Since Loan_Burden (loan size relative to income, combined with DTI) is the stronger and more consistent driver of default found in this analysis, make it — not raw DTI — the primary debt-burden underwriting gate.
- Introduce a hard affordability floor using **Income_After_EMI**: applications that model out to negative disposable income after the proposed EMI should require explicit exception approval rather than passing through standard underwriting, directly targeting the >25% of the book currently carrying this exposure.

4.3 Set a specific EMI-affordability cap rather than relying on book-wide averages

- Report EMI-to-income and Income-After-EMI as **distribution views (e.g. share of book beyond a threshold)** in the dashboards, not just as a book-wide average, so the unaffordable tail stays visible to decision-makers rather than being averaged away.
- Set a maximum EMI-to-income threshold for standard approval (e.g. flag anything materially above 1.0× monthly income) and route applicants above it into the same enhanced-review path as model-flagged applications.

4.4 Turn the known high-risk combinations into explicit underwriting rules

- **Require a co-signer above a defined loan-size threshold** — the data shows this single rule would move the riskiest segment (no co-signer, high loan amount, 15.42% default) toward the safest segment's profile (co-signer, low loan amount, 8.20%).
- Apply additional scrutiny or tighter terms specifically to the **Unemployed + Business-loan-purpose** combination, the widest risk spread found in the data (14.37% vs. 8.05% for the safest combination), rather than relying on employment type or loan purpose individually.

4.5 Build a portfolio-health layer alongside the existing growth dashboards

- Add default rate, 30/60/90-day delinquency buckets, expected credit loss, portfolio-at-risk, and recovery rate as standing KPIs reported at the same cadence and prominence as total disbursed and approval rate, so growth and risk are always read together.
- Use the model's predicted-probability output as a forward-looking **expected default rate** estimate for the current pipeline of applications, giving management an early read on portfolio quality before those loans season.

4.6 Investigate and act on the post-May application decline

- Attribute the June-onward volume drop to a specific cause — marketing/channel spend, a policy or eligibility change, or genuine seasonal demand — by checking it against internal marketing spend and policy-change records for the same period.
 - If the cause is a controllable one (campaign end, tightened eligibility), decide deliberately whether to restore volume or hold the tighter stance, rather than treating the lower volume as an unexplained given.
-

5. Conclusion

This project delivers a working, recall-oriented default-prediction model — deliberately tuned to catch likely defaulters even at the cost of some false alarms — ready to sit inside the underwriting workflow as a screening layer. Read alongside the institution's own dashboards, the case for using it is strong: current approval behavior is running ahead of the applicant pool's actual risk profile, the debt-burden and affordability signals that most reliably predict default are not yet being used as hard gates, and portfolio risk currently has no dedicated reporting layer at all. None of the fixes proposed above require new data collection — they use features and a model already built in this project. The highest-leverage next step is operational, not technical: wire the model's output and the `Loan_Burden / Income_After_EMI` thresholds into the live underwriting decision, and put default-rate and delinquency reporting next to the existing growth dashboards.

Prepared as an internal business review based on the project's EDA, modeling results, and executive dashboard analyses.